

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q19-1181

Dresden Generating Station 100 kV  
April 24, 2021

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1181. The Interconnection Customer proposes a 205.0 MW solar photovoltaic facility interconnecting at the Dresden Generating Station 100 kV substation in the Illinois area. The request was received on April 04, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 298 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$174,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

MORRIS COGENERATION LLC – BOCA RATON 230.0 kV L: 102.9% loading (676 MVA) following the outage of the DRESDEN GENERATING STA – MORRIS COGENERATION LLC 230.0 kV; DFAX = 7.0%.

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                     | Required Upgrade           | Cost Estimate | Allocation to Project |
|----------------------------------------------|----------------------------|---------------|-----------------------|
| BOC RATON – MORRIS COGENERATION LLC 230.0 kV | Reconductor line           | \$6,727,000   | 3% (\$174,000)        |
|                                              | Total allocated to project |               | \$174,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.